Orbit unions and exchange walks: seven new lower bounds for binary codes

Zhiyang Chen Ligeng Zhu Fan Long

October 2026

Abstract

Let $A(n, d)$ be the largest size of a binary code of length n and minimum distance d. We give explicit codes that show $A(17, 6) \geq 260$, $A(21, 6) \geq 3072$, $A(24, 10) \geq 144$, $A(25, 10) \geq 256$, $A(27, 6) \geq 81920$, $A(28, 6) \geq 139264$ and $A(28, 12) \geq 192$. Six of the codes are unions of orbits of a group of isometries: we choose the group, build the graph of orbits, and select orbits by a maximum independent set search. Most of the groups are translation groups of $\mathbb{Z}_4$-linear codes under the Gray map. The seventh code was found by an exact exchange walk that starts from such a code. Each code is given as a file, and its minimum distance is checked directly.

1 Results

Table 1 lists the bounds; "old" is the best value known to us before this work. Four of the bounds were sent to A. E. Brouwer on 2026-10-21 and are now in his table [1]. Since $A(n-1, d) \geq A(n, d)/2$ (shortening), we also get $A(26, 6) \geq 40960$ and $A(25, 6) \geq 20480$ (old 32768 and 17920).

(n, d)	old	new	construction
$(17, 6)$	258 [5]	**260**	exchange walk from an orbit union
$(21, 6)$	2560	**3072**	48 cosets of a $\mathbb{Z}_4$-code of order 64, with 4 extra coordinates
$(24, 10)$	136 [4]	**144**	18 cosets of a 3-dimensional subcode of the Golay code
$(25, 10)$	196 [6]	**256**	16 cosets of a $\mathbb{Z}_4$-code of order 16, with 3 extra coordinates
$(27, 6)$	65536	**81920**	20 cosets of a $\mathbb{Z}_4$-code of order 4096, with 3 extra coordinates
$(28, 6)$	131072	**139264**	34 cosets of the same $\mathbb{Z}_4$-code, with 4 extra coordinates
$(28, 12)$	178	**192**	12 cosets of a 4-dimensional binary linear code

Table 1: New lower bounds. The bounds for $(21, 6)$, $(27, 6)$, $(28, 6)$ and $(28, 12)$ are already in [1].

2 The method

Orbit unions. Let Γ be a group of isometries of $(\mathbb{F}_2^n, d_H)$. A union of Γ-orbits $O_1 \cup \cdots \cup O_m$ has minimum distance at least d if and only if each O_i has internal minimum distance at least d and $d(O_i, O_j) \geq d$ for $i \neq j$. Since Γ acts by isometries, $d(O, O') = \min_{y \in O'} d(x, y)$ for any fixed $x \in O$. So the largest Γ-invariant code is a maximum-weight independent set in the graph whose vertices are the orbits with internal distance $\geq d$ (weight $|O|$), two orbits being adjacent when their distance is $< d$. We choose Γ so that this graph is small, and search it with tabu search, the CHILS heuristic [2] or an exact clique solver. We use three kinds of groups:

- *Translations by a binary linear code L.* The orbits are the cosets $x + L$; the internal distance is the minimum weight of L, and $d(x + L, y + L)$ is the minimum weight of $x + y + L$.
- *Translations by a $\mathbb{Z}_4$-linear code $K \le \mathbb{Z}_4^m$.* The Gray map $\phi : \mathbb{Z}_4 \to \mathbb{F}_2^2$, $0 \mapsto 00$, $1 \mapsto 01$, $2 \mapsto 11$, $3 \mapsto 10$, is an isometry from the Lee metric to the Hamming metric [3]. So the maps $\phi(x) \mapsto \phi(x + k)$, $k \in K$, are isometries of $\mathbb{F}_2^{2m}$. We also add e binary coordinates, i.e. we work in $\mathbb{F}_2^e \times \mathbb{Z}_4^m$ with the distance $d_H + d_{\mathrm{Lee}}$, and take unions of cosets of $\{0\} \times K$. A coset has internal distance equal to the minimum Lee weight of K.
- *Coordinate permutations*, e.g. a cyclic group generated by one permutation.

A useful choice for K is a subgroup of small index in a $\mathbb{Z}_4$-linear code of large minimum Lee distance: its cosets inside that code are pairwise compatible, and the extra coordinates and the cosets outside the code give room for more.

Exact exchange walks. Let C be a code with minimum distance d. A move removes a set R of $k \le 60$ codewords near a random point, and computes (exactly, with a node limit) a largest set R' of words that are at distance $\ge d$ from $C \setminus R$ and from each other. If $|R'| \ge |R|$ the move is accepted. So the walk moves on the plateau of codes of equal size until a move gives $|R'| > |R|$. Codes from one large group are usually rigid (no equal-size move exists), so we start walks from many structurally different codes of the best known size or slightly below it, and every code that a walk grows becomes a new start. In practice we puncture a random coordinate, walk on the $(n - 1, M, d - 1)$ code, and add the parity bit back; a node limit makes a small fraction of the searches inexact. This affects only the search: the final code is checked directly.

3 The codes

For even d, an overall parity bit turns an $(n - 1, M, d - 1)$ code into an (n, M, d) code ("parity extension"). In the files of the $\mathbb{Z}_4$ constructions the e extra binary coordinates come first, followed by the Gray images; the only exception is $A(21, 6)$, where the 16 Gray coordinates come first, then the 4 extra coordinates and the parity bit. The coset lists and the formats are in the data files.

$A(17, 6) \ge 260$. Let $\sigma = (0\,1\,2\,3\,4)(5\,6\,7\,8\,9)(10\,11\,12\,13\,14)$ act on 16 coordinates. CHILS on the graph of $\langle\sigma\rangle$-orbits gave a $\langle\sigma\rangle$-invariant $(16, 257, 5)$ code; its parity extension is a $(17, 257, 6)$ code. An exchange walk on it reached a $(17, 259, 6)$ code within a walk of 1.3 million moves, and a second walk from that code reached a $(17, 260, 6)$ code after 0.77 million moves. The final code has no symmetry that we imposed; its kernel $\{v : v + C = C\}$ is trivial.

$A(21, 6) \ge 3072$. Let $O \le \mathbb{Z}_4^8$ be the octacode ($|O| = 256$, minimum Lee distance 6; its Gray image is the Nordstrom–Robinson code [3]). For a subgroup $H \le O$ of order 64, a union of 48 cosets of $\{0\} \times H$ in $\mathbb{F}_2^4 \times \mathbb{Z}_4^8$ is a $(20, 3072, 5)$ code, and its parity extension is a $(21, 3072, 6)$ code. (The best union of cosets of the full octacode in this setting has only 2560 words.)

$A(24, 10) \ge 144$. Let L be a 3-dimensional subcode of the binary Golay code $[23, 12, 7]$ with minimum weight 11. A union of 18 cosets of L in $\mathbb{F}_2^{23}$ is a $(23, 144, 9)$ code (the cosets need not lie in the Golay code; here none does); its parity extension is a $(24, 144, 10)$ code. It is not linear, since 144 is not a power of 2.

$A(25,10) \geq 256$**.** Let $C \leq \mathbb{Z}_4^{11}$ be generated by

$$g_1 = (2,2,2,0,2,0,0,0,2,0,0), \quad g_2 = (2,3,3,1,0,0,1,2,1,1,0), \quad g_3 = (1,0,2,1,2,1,3,3,0,0,1).$$

Then $|C| = 32$ and $C = \langle g_1 \rangle \oplus \langle g_2 \rangle \oplus \langle g_3 \rangle$ has minimum Lee distance 10. Let $K = \{ag_1 + bg_2 + cg_3 : a + b + c \text{ even}\}$, of order 16 and minimum Lee distance 10. A union of 16 cosets of $\{0\} \times K$ in $\mathbb{F}_2^3 \times \mathbb{Z}_4^{11}$ is a $(25, 256, 10)$ code.

$A(27,6) \geq 81920$ **and** $A(28,6) \geq 139264$**.** Let $C = \ker \Phi \leq \mathbb{Z}_4^{12}$, where $\Phi : \mathbb{Z}_4^{12} \to \mathbb{Z}_2^2 \times \mathbb{Z}_4^4$ is the homomorphism given in the data file `z12_1.h`; then $|C| = 2^{14}$ and C has minimum Lee distance 6. Let $\psi : \mathbb{Z}_4^{12} \to \mathbb{Z}_2^2$, $\psi(x) = \sum_{i:\, x_i \text{ odd}} \psi_i$, with $(\psi_1, \ldots, \psi_{12}) = (1,2,1,1,2,1,3,0,0,0,2,1)$ (an integer $t \in \{0,1,2,3\}$ stands for its binary digits). Then $C' = C \cap \ker \psi$ has order 4096. A union of 20 cosets of $\{0\} \times C'$ in $\mathbb{F}_2^3 \times \mathbb{Z}_4^{12}$ is a $(27, 81920, 6)$ code, and a union of 34 cosets in $\mathbb{F}_2^4 \times \mathbb{Z}_4^{12}$ is a $(28, 139264, 6)$ code. Both rebuild exactly from (Φ, ψ) and the coset lists in the data files.

$A(28,12) \geq 192$**.** The code is a union of 12 cosets of a 4-dimensional linear code $L \leq \mathbb{F}_2^{28}$ of minimum weight 12, and it lies in a coset of a 12-dimensional linear code. It was found by a coset search with a smaller kernel taken inside a good linear code of length 27, followed by parity extension.

4 Verification and data

Each code is a list of binary words. Its size, its length and the absence of two words at distance $< d$ were checked by at least two independent programs; this finite check is the proof. The structure stated above was also checked: the $\mathbb{Z}_4$ codes were rebuilt from their parameters, and the cosets and kernels were recomputed from the files.

file	SHA-256 (first 16 hex digits)
`A_17_6_260.txt`	`489b6f11abdc0ae7`
`A_21_6_3072.txt`	`dfc23bec786e9375`
`A_24_10_144.txt`	`d723cfa082ea9230`
`A_25_10_256.txt`	`8c64ab1860a01dd8`
`A_27_6_81920.txt`	`0038594ad6dd0b20`
`A_28_6_139264.txt`	`df5c845a6c748619`
`A_28_12_192.txt`	`5149782b3253fe49`

The code files, the group and coset data, and the search programs are in the data record of this paper (`https://doi.org/10.5281/zenodo.23599650`).

Use of AI. AI tools were used in the search and in the writing of this paper. The correctness of the bounds does not depend on them: each code is checked directly, as described above, and the authors guarantee the stated bounds.

References

[1] A. E. Brouwer, Table of general binary codes, `https://aeb.win.tue.nl/codes/binary-1.html`.

[2] E. Großmann, K. Langedal, C. Schulz, Accelerating reductions using graph neural networks and a new concurrent local search for the maximum weight independent set problem, arXiv:2412.14198.

[3] A. R. Hammons, P. V. Kumar, A. R. Calderbank, N. J. A. Sloane, P. Solé, The $\mathbb{Z}_4$-linearity of Kerdock, Preparata, Goethals, and related codes, *IEEE Trans. Inform. Theory* 40 (1994) 301–319.

[4] A. Laaksonen, P. R. J. Östergård, Constructing error-correcting binary codes using transitive permutation groups, *Discrete Appl. Math.* 233 (2017) 65–70.

[5] M. Milshtein, A new two-error-correcting binary code of length 16, *Cryptogr. Commun.* (2019).

[6] K. Ota, T. Osa, T. Harada, Three improved lower bounds for binary error-correcting codes, arXiv:2610.11609 (2026).